

K L Deemed to be University

Department of Computer Science and Engineering | 2026-2027, Odd Semester
Database Systems Engineering and Distributed Backend Development (25CS1302E)

Practical Experiment 1: The Foundation (BookFlow SQL Schema)

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Aim

To build the SQL foundation of the BookFlow library system: create the database bookflow_db, create the Books and Members tables with PRIMARY KEY, NOT NULL, UNIQUE and CHECK constraints (unique ISBN, valid email, published_year < 2027), seed three books and three members, and verify that the rules reject invalid data.

Program

p1.sql (run with: psql -U postgres -f p1.sql)

```
-- Experiment 1: The Foundation - BookFlow SQL schema
-- Step 1: Database setup
DROP DATABASE IF EXISTS bookflow_db;
CREATE DATABASE bookflow_db;
\c bookflow_db

-- Step 2 and 3: Table creation (DDL) with constraints
CREATE TABLE Books (
    book_id      SERIAL      PRIMARY KEY,
    title        VARCHAR(200) NOT NULL,
    isbn         VARCHAR(13)  NOT NULL UNIQUE,
    published_year INT        CHECK (published_year < 2027)
);

CREATE TABLE Members (
    member_id SERIAL      PRIMARY KEY,
    full_name  VARCHAR(100) NOT NULL,
    email      VARCHAR(150) NOT NULL UNIQUE
    CHECK (email ~* '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$')
);

\d Books
\d Members

-- Step 4: Initial seeding (3 books, 3 members)
INSERT INTO Books (title, isbn, published_year) VALUES
('Clean Code', '9780132350884', 2008),
('Database System Concepts', '9780078022159', 2019),
('The Martian', '9780553418026', 2014);

INSERT INTO Members (full_name, email) VALUES
('Habeeb', 'habeeb@kluniversity.in'),
('Ananya Rao', 'ananya.rao@gmail.com'),
('Rahul Verma', 'rahul.v@outlook.com');

SELECT * FROM Books;
SELECT * FROM Members;

-- Verifying that the rules work (each statement below must be rejected)
-- 1. Duplicate ISBN
INSERT INTO Books (title, isbn, published_year) VALUES ('Copy of Clean Code', '9780132350884', 2010);
-- 2. Book without a title
INSERT INTO Books (title, isbn, published_year) VALUES (NULL, '9781234567897', 2020);
-- 3. Published year in the future
INSERT INTO Books (title, isbn, published_year) VALUES ('Future Book', '9789999999999', 2030);
-- 4. Member without an email
INSERT INTO Members (full_name, email) VALUES ('No Email', NULL);
-- 5. Member with an invalid email
INSERT INTO Members (full_name, email) VALUES ('Bad Email', 'not-an-email');
-- 6. Duplicate email
INSERT INTO Members (full_name, email) VALUES ('Duplicate', 'habeeb@kluniversity.in');
```

```
SELECT COUNT(*) AS total_books FROM Books;
SELECT COUNT(*) AS total_members FROM Members;
```

Output

```
DROP DATABASE IF EXISTS bookflow_db;
DROP DATABASE
```

```
CREATE DATABASE bookflow_db;
CREATE DATABASE
```

```
CREATE TABLE Books (
    book_id SERIAL PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    isbn VARCHAR(13) NOT NULL UNIQUE,
    published_year INT CHECK (published_year < 2027)
);
CREATE TABLE
CREATE TABLE Members (
    member_id SERIAL PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE
    CHECK (email ~* '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$')
);
CREATE TABLE
```

Column	Type	Table "public.books"	Collation	Nullable	Default
book_id	integer			not null	nextval('books_book_id_seq'::regclass)
title	character varying(200)			not null	
isbn	character varying(13)			not null	
published_year	integer				

Indexes:

"books_pkey" PRIMARY KEY, btree (book_id)

"books_isbn_key" UNIQUE CONSTRAINT, btree (isbn)

Check constraints:

"books_published_year_check" CHECK (published_year < 2027)

Column	Type	Table "public.members"	Collation	Nullable	Default
member_id	integer			not null	nextval('members_member_id_seq'::regclass)
full_name	character varying(100)			not null	
email	character varying(150)			not null	

Indexes:

"members_pkey" PRIMARY KEY, btree (member_id)

"members_email_key" UNIQUE CONSTRAINT, btree (email)

Check constraints:

"members_email_check" CHECK (email::text ~* '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\$'::text)

```
INSERT INTO Books (title, isbn, published_year) VALUES
```

```
('Clean Code', '9780132350884', 2008),
```

```
('Database System Concepts', '9780078022159', 2019),
```

```
('The Martian', '9780553418026', 2014);
```

```
INSERT 0 3
```

```
INSERT INTO Members (full_name, email) VALUES
```

```
('Habeeb', 'habeeb@kluniversity.in'),
```

```
('Ananya Rao', 'ananya.rao@gmail.com'),
```

```
('Rahul Verma', 'rahul.v@outlook.com');
```

```
INSERT 0 3
```

```
SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	Clean Code	9780132350884	2008
2	Database System Concepts	9780078022159	2019
3	The Martian	9780553418026	2014

(3 rows)

```
SELECT * FROM Members;
```

member_id	full_name	email
1	Habeeb	habeeb@kluniversity.in
2	Ananya Rao	ananya.rao@gmail.com
3	Rahul Verma	rahul.v@outlook.com

(3 rows)

```
INSERT INTO Books (title, isbn, published_year) VALUES ('Copy of Clean Code', '9780132350884', 2010);
```

ERROR: duplicate key value violates unique constraint "books_isbn_key"

DETAIL: Key (isbn)=(9780132350884) already exists.

```
INSERT INTO Books (title, isbn, published_year) VALUES (NULL, '9781234567897', 2020);
```

```
ERROR: null value in column "title" of relation "books" violates not-null constraint
DETAIL: Failing row contains (5, null, 9781234567897, 2020).
INSERT INTO Books (title, isbn, published_year) VALUES ('Future Book', '9789999999999', 2030);
ERROR: new row for relation "books" violates check constraint "books_published_year_check"
DETAIL: Failing row contains (6, Future Book, 9789999999999, 2030).
INSERT INTO Members (full_name, email) VALUES ('No Email', NULL);
ERROR: null value in column "email" of relation "members" violates not-null constraint
DETAIL: Failing row contains (4, No Email, null).
INSERT INTO Members (full_name, email) VALUES ('Bad Email', 'not-an-email');
ERROR: new row for relation "members" violates check constraint "members_email_check"
DETAIL: Failing row contains (5, Bad Email, not-an-email).
INSERT INTO Members (full_name, email) VALUES ('Duplicate', 'habeeb@kluniversity.in');
ERROR: duplicate key value violates unique constraint "members_email_key"
DETAIL: Key (email)=(habeeb@kluniversity.in) already exists.
SELECT COUNT(*) AS total_books FROM Books;
total_books
-----
3
(1 row)

SELECT COUNT(*) AS total_members FROM Members;
total_members
-----
3
(1 row)
```

Result

bookflow_db was created with the Books and Members tables. Three books and three members were inserted, and all six invalid inserts were rejected: duplicate ISBN and duplicate email by UNIQUE, missing title and missing email by NOT NULL, and the year 2030 and the malformed email by CHECK constraints. The tables still contain exactly 3 books and 3 members.